The DICTO Palintriacontahedron

The rhombic triacontahedron turned inside out: the hull of the icosidodecahedron star at full height, its icosahedron edge 3/√5 times its dodecahedron edge

DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com
Version 1 · 17 October 2026 · DICTO DOI: on publication · Licence CC BY 4.0 · An introduction to one solid; its parent paper: Vajrayana, DICTO DOI: 10.5281/zenodo.23429911. Data and checks are included with this record.
Flipping the icosidodecahedron's own central pyramids out at full height ("shortening feels like cheating; one more solid is needed") and the name are DICTO's; the exact computations and the text were done with an AI assistant (Claude, Anthropic).

Abstract. Flip the icosidodecahedron's 32 central pyramids out at full height and take the hull of their tips: the result is the DICTO Palintriacontahedron X (Greek *pálin*, back), with 32 corners and 60 triangular faces, each half a rhombic-dodecahedron rhombus ($\cos^2$ of its angles 1/9, 1/3, 1/3). It is the member of the dodecahedron–icosahedron hull family whose icosahedron edge is exactly 3/√5 times its dodecahedron edge, in closed form the hull of (0, ±1, ±φ) and ((5 + √5)/6)·{(±1, ±1, ±1), (0, ±φ, ±1/φ)} (cyclic). It has the rhombic triacontahedron's 32 axes with the reciprocal radius ratio: the triacontahedron turned inside out. Everything here is exact in ℚ(√5) unless marked float.

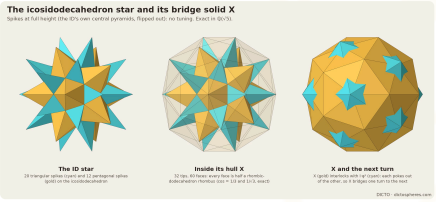


Figure 1. The icosidodecahedron star (its central pyramids flipped out at full height), inside its hull, the DICTO Palintriacontahedron, and X interlocked with the next icosahedron. Drawn from exact corners.

1. The solid

With the icosahedron (0, ±1, ±φ) (cyclic, edge 2) and the dodecahedron (±1, ±1, ±1), (0, ±φ, ±1/φ) (cyclic) placed over its faces, the hull of the icosahedron and k times the dodecahedron has edge ratio φ/k. The DICTO Palintriacontahedron is k = φ√5/3 = 5/6 + √5/6, ratio 3/√5 (squared 9/5).

Property	Value (exact)
corners	32: 12 five-fold, 20 three-fold, all tips of the full-height icosidodecahedron star
faces, edges	60 triangles, each half a rhombic-dodecahedron rhombus ($\cos^2$ of the face angles 1/9, 1/3, 1/3); 90 edges
squared edge lengths	20/9 (60 edges) and 5/3 (30 edges)
squared corner radii	5/2 + √5/2 (five-fold) and 5/2 + 5√5/6 (three-fold)
insphere	all 60 faces at squared distance 7/4 + 3√5/4
midsphere	none (edges at squared distances 25/12 + 2√5/3 and 35/18 + 5√5/6)
volume	50/3 + 50√5/9 ≈ 29.0893 (in these units)

Table 1. The DICTO Palintriacontahedron in closed form (`scripts/hull_family_exact.py`, exact).

2. Where it comes from

In the twelve-station golden cycle of Vajrayana (DICTO DOI: 10.5281/zenodo.23429911) the icosidodecahedron is the shell where the cycle's two progenitors meet. DICTO flipped its central pyramids outward at full height rather than shortening them; the hull of the 32 tips is X. It is the rhombic triacontahedron turned inside out: the same 32 axes, with the five-fold and three-fold corners at the reciprocal ratio of radii. Two float observations (labelled as such in Vajrayana): X lies inside the next turn's D′ and 99.46% of its volume (20,000 sample points) lies inside the next icosahedron's Cleo star, so it bridges one turn to the next; and the first stellation of the icosahedron puts its tips at 3/√5 times the face distance, the same number.

3. Its family

X belongs to the same hull family as the pentakis dodecahedron (Catalan, ratio $3(7 + 5\sqrt{5})/38 \approx 1.435$) and the DICTO Metaxyhexecontahedron ($9/\varphi^4 \approx 1.313$), whose introduction accompanies this one. Its ratio $3/\sqrt{5} \approx 1.342$ lies between them.

4. Status

Known: the hull family and its fully symmetric range (Polytope Wiki), the rhombic triacontahedron and the icosidodecahedron's central pyramids. **Not found, in the sources checked** (Vajrayana §5, SEARCH-LOG.md): this member as a named or distinguished solid, or the ratio 3/√5 as distinguished. The name returns no results. Every "not found" is limited to those sources.

5. Files

`scripts/hull_family_exact.py`, `data-hull-family-exact.txt` (this paper, exact); `id_star.py` in the study (branch research: studies/2026-10-16-progenitors). See it in the app: Kaleidohedra, 3D+ → Vajrayana.